

Dr. Babasaheb Ambedkar Technological University, Lonere

B.Tech-Electrical Engineering

Semester:-VIII

Subject: - (EE801) Electrical Utilization (SI)

Summer Semester

Time: 3 hour

Date-10/11/2016
10/11/2016

Marks: 70

Q.1	I) The process of smoothing out of fluctuations in load is known as ---- a) load equalization b) steady state stability c) either a or b d) none of these II. -----cells are based on cell reactions which are reversible. a. PV b. Secondary c. Primary d. Both a & c III. Luminous efficiency of Tungsten filament lamps increases with increase in..... a) Power b) Voltage c) Current d) Wattage IV. Infra-red heating is also known as a) Induction heating b) Radiant heating c) resistance heating d) either a or b V. Candela is the unit of ---- a. Illumination b) luminous intensity c) luminous flux d) either b or c VI. Slipping will not take place unless tractive effort F---- a) $=\mu W$ b) $< \mu W$ c) $> \mu W$ d) none of the above VII. The ratio of tractive effort to slip the wheels to the Adhesive weight is a) Coefficient of adhesion b) Coefficient of friction c) Gradient d) either a & c VIII. At equilibrium position, the load torque is equal to---- a) Friction torque b) motor torque c) Inertia torque d) dynamic torque IX. The energy output at the driving axels is spent to : a) Accelerate the train B) Overcome the gradient c) Both a & b d) Neither a nor b X. Total weight to be carried on the driving wheels is known as----- a) Dead weight b) Accelerating weight c) Decelerating weight d) None of these	10M
Q.2	a) Explain with the neat sketch construction & working of lead-acid battery. b) A drive has following parameters; $J=10\text{kg-m}^2$, $T=100-0.1N$, N-m, passive load torque $T_L=0.05N$, N-m where N is the speed in rpm. Now it is to be reversed. For this motor characteristic is changed to $T=100-0.1N\text{-m}$.	6M

